Separating Latent SLCI Test-Taking Groups *

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Executive Summary

This report aims to validate the existence of two distinct latent groups of respondents within the population taking the Science Literacy Concept Inventory (SLCI). Establishing these groups and accurately characterizing their performance is important for separating genuine lack of knowledge from low engagement or guessing behavior. In Section 1, we outline the primary research questions regarding the distribution of SLCI scores and the potential influence of student demographics. In Section 2, we describe the dataset structure, focusing on the simplification of academic rank categories and the handling of missing data to ensure a homogeneous undergraduate sample. In Section 3, we conduct an exploratory data analysis, examining score distributions across key covariates such as academic rank, native English, and science course completion, while noting the lack of a clear bimodal distribution in raw scores. In Section 4, we introduce the first Bayesian Beta-Binomial mixture model, which establishes a baseline separation of the two groups without demographic inputs. In Section 5, we expand this model to include demographic covariates, revealing how student background characteristics significantly shift the probability of group membership. Finally, Section 6 refines the analysis with a fully heterogeneous mixture model that allows for variable performance within the engaged group, successfully isolating a distinct low engagement cluster scoring near chance and providing evidence that engagement is influenced by academic demographics. Key findings and design recommendations for future assessments are summarized in Section 8, with supplementary diagnostics and figures provided in the Appendix.

(1) Introduction

The goal of this report is to provide evidence regarding the existence of two distinct groups of participants who took the validated Science Literacy Concept Inventory (SLCI) and postdicted global self-assessment: random or low engagement respondents (**R**) and engaged respondents (**I**).

(2) Data

The variable of interest in this analysis is `SLCI Score Count` (the number of correct answers), with the additional covariates of `Academic Rank`, `Science Courses`, `Intent`, and `English`.

Simplifying Academic Rank

To use the proposed covariates, we need to check the counts at each level of the covariates:

Table 1: Counts by Academic Rank

Freshman	Grad Student	HS Senior	Junior	Practitioner	Professor	Senior	Sophomore	Sum
4104	262	19	1800	9	213	1769	2991	11167

*Formatting based on svmille guide (http://github.com/svmiller).

Table 2: Counts by Completed Science Courses

None	One	Two	Three	Four	More than four	Sum
3081	1970	1589	1072	563	2954	11229

Table 3: Counts by English as First Language

No	Yes	Sum
1312	9842	11154

The low counts in the `Academic Rank` levels *Grad Student*, *HS Senior*, *Practitioner*, and *Professor* motivate removing these records from the analysis to prevent sparse categories and subsequent unstable parameter estimates. In doing so, the population studied is narrowed to just undergraduate students, which is a more homogeneous group in terms of assessment stakes and test-taking behavior.

Missingness

To check the missing values in the data, we create a table showing how many NA values remain in the data after filter to just undergraduate students:

Table 4: Counts by Science Courses

None	One	Two	Three	Four	More than four	NA	Sum
2978	1933	1549	1031	543	2614	16	10664

Table 5: Counts by English as First Language

Yes	No	NA	Sum
9386	1249	29	10664

Table 6: Counts by Intent to or Major in Science

No	Yes	NA	Sum
4170	6469	25	10664

Only $16 + 29 + 25 = 70$ NA observations, which is too few to suggest any patterns compared to the thousands of complete records. Therefore, we can just drop these records with missing observations.

(3) Exploratory Data Analysis

Before fitting any models, we need to check to see how this subset of data relates to the expected result.

Score Distributions

Figure 1 shows the SLCI Score Count distribution among undergraduate students. The most important consideration here is determining whether a proportion of students appear to be guessing. On a 25 question test where each question has four options, the expected SLCI Score Count of a student guessing would theoretically be $0.25 \times 25 = 6.25$. For a Binomial$(n = 25, p = 0.25)$ we would expect 80% of random guessers to have a SLCI Score Count between 4 and 9. The percentage of students who scored in this range was 6.5% of all test-takers. While it might be tempting to include scores below 4 in this group, the comedic nature of the questions likely points to an even smaller sub-population who is answering questions ironically.

The distribution shown in Figure 1 shows only a small bump near the 6.25 area, which is insufficient to suggest a bimodal population. This lack of separation makes it difficult to distinguish random guessers from low-performing students, so we keep the **R** group definition (guessers <u>or</u> low-engaged) to also encompass those scoring barely above chance and those answering ironically. Furthermore, this motivates the inclusion of other covariates in any classification models because latent groups may overlap heavily and covariates can help resolve this ambiguity.

Figure 2 shows the distribution of SLCI Score Count across different Academic Rank levels. Although the interquartile ranges overlap across all levels, the notches indicate a statistically significant difference in the median SLCI Score Count for all ranks except between *Freshman* and *Sophomore*.

Similarly, Figure 3 shows a significant difference in the median SLCI Score Count between undergraduates who reported English as their first language (*Yes*) and those who did not (*No*).

The distribution of SLCI Score Count across the levels of Science Courses taken, shown in Figure 4, is less distinct. The notches in the boxplots show no significant difference in the median SLCI Score Count between levels *One* and *Two*, and between levels *Three* and *Four*. Also, scoring variation appears to decrease (indicating greater reliability) as the number of Science Courses taken increases.

Finally, the covariate Intent (intent to major or majoring in science) also shows a significant difference in the median SLCI Score Count between its two levels, *Yes* and *No*, as seen in Figure 5.

This preliminary analysis suggests that all selected covariates will be valuable for accurately modeling the score distribution of undergraduate students, given the significantly different medians observed across nearly all levels of these variables.

Heteroscedasticity in Self-Assessment Inaccuracy

One consideration is the heteroscedasticity observed in Figure 6, which displays Self-Assessment Inaccuracy (Postdicted Rating $-$ SLCI Score Count) across bins of SLCI Score Count (expressed as a percentage). While this variance would be problematic if the goal were to define guessers based on their self-assessment errors, we instead opt to exclude Self-Assessment Inaccuracy from the model, given that the Dunning-Kruger effect is questioned in the associated paper. Our models will instead rely solely on a student's SLCI Score Count and the previously discussed demographic covariates to predict group **R** membership. Consequently, the model is only required to handle the heteroscedasticity inherent in the SLCI Score Count itself, which will be discussed further during model building.

(4) Approach One

The first approach is to ignore covariates for now, and build a Bayesian model just based on SLCI Score Count .

Model 1 Notation

Define the observed outcome (number of correct answers out of 25 for student i) as:

$$X_i \in \{0, \ldots, 25\}$$

For a given student, imagine they have a latent per-question correctness probability p_i. Then:

$$X_i \mid p_i \sim \text{Binomial}(25, p_i)$$

Since students differ in factors like knowledge and engagement, we assign a Beta prior to p_i:

$$p_i \sim \text{Beta}(\alpha, \beta)$$

Here, we rewrite these parameters for interchangeability as $\mu = \frac{\alpha}{\alpha+\beta}$ and $\phi = \alpha + \beta$, representing location and scale parameters of the Beta distribution.

Marginalizing over p_i gives the Beta-Binomial distribution:

$$X_i \sim \text{Beta-Binomial}(25, \mu, \phi)$$

We assume the two latent groups previously discussed exist: random or low engagement respondents (**R**) and engaged respondents (**I**). Let $z_i \in \{R, I\}$ be the latent group. We model the SLCI Score Count distribution as a mixture:

$$P(X = x) = w \cdot \text{BB}(x \mid \mu_R, \phi_R) + (1 - w) \cdot \text{BB}(x \mid \mu_I, \phi_I)$$

Here w is the population proportion of random or low engagement students (corresponding to $1 - \theta_2$ in the model estimation). Once this mixture is fitted, we will have the density function to compute the probability a SLCI Score Count x being in group **R**:

$$f_R(x) = P(R \mid X = x) = \frac{w \cdot \text{BB}(x \mid \mu_R, \phi_R)}{w \cdot \text{BB}(x \mid \mu_R, \phi_R) + (1 - w) \cdot \text{BB}(x \mid \mu_I, \phi_I)}$$

We set a strong prior for μ_R near ≈ 0.27 and a high precision (ϕ_R) prior so it stays close to a guess-like location rather than just a low ability location.

Model 1 Results

The package we use to fit the Bayesian regression is *brms* (this package is used for all models in this report). As a reminder, this uses (μ, ϕ) for each group:

$$\mu = \frac{\alpha}{\alpha + \beta}, \quad \phi = \alpha + \beta$$

$$\alpha = \mu\phi, \quad \beta = (1 - \mu)\phi$$

The model successfully converged ($\hat{R} = 1.00$ for all parameters), identifying two latent groups with distinct performance characteristics summarized in the model output.

1. The low engagement group (**R**)

The first group, intended to capture random guessers, estimated a mean success probability of $\mu_R \approx 0.35$ (from the logit intercept of -0.60).

- Implied score: This corresponds to an average score of 8.87 out of 25.
- Deviation from hypothesis: This result is notably higher than the theoretical random guessing baseline of 6.25 (25%) and the prior we set at ≈ 6.75 (27%). The data provided strong enough evidence to pull the posterior away from the rigid guessing prior.
- Interpretation: This suggests that the lowest performing cluster of students is not responding purely at random. Instead, they possess partial knowledge or low engagement that results in scores slightly better than chance.

2. The engaged group (**I**)

The second group captured the majority of the student population with a mean success probability of $\mu_I \approx 0.75$ (logit intercept 1.12).

- Implied score: This corresponds to an average score of 18.86 out of 25 (75.4%).
- Precision: The engaged group exhibits greater variability in performance, consistent with a heterogeneous population of genuinely attempting students.

3. Population proportions

The mixing parameter θ_2 (intercept 1.74) estimates the proportion of students belonging to the engaged group.

- Converting the logit estimate: $\text{logit}^{-1}(1.74) \approx 0.85$.
- Result: Approximately 85% of the population falls into the high-performing engaged (**I**) group, while roughly 15% are classified into the low engagement (**R**) group.

(5) Approach Two

Next, we include the additional demographic covariates in the model to help separate the population proportions of the two latent groups.

Model 2 Notation

The simplest way to include extra covariates is to have them affect the mixing:

$$\theta_{2,i} = P(z_i = I \mid \cdot) = \text{logit}^{-1}(\eta_i)$$

$$w_i = P(z_i = R) = 1 - \theta_{2,i}$$

We make the Academic Rank and Science Courses be random effects in the model, so that each factor contributes one variance parameter (σ_{year} and σ_{sci}). That way, if a variance is near zero, that factor can be ignored without having to do multiple testing. The linear predictor from the logit is:

$$\eta_i = \gamma_0 + u_{\text{year}[i]} + v_{\text{sci}[i]} + \beta_E \cdot 1\{\text{English}_i = \text{No}\}$$

With the hierarchical priors:

$$u_j \sim N(0, \sigma_{\text{year}}), \quad v_k \sim N(0, \sigma_{\text{sci}})$$

We assign weakly informative priors to the parameters controlling the group-level variances, encouraging the model to shrink the random effects toward zero unless the data demands otherwise:

$$\sigma_{\text{year}}, \sigma_{\text{sci}} \sim \text{Exponential}(1)$$

For the fixed effects determining the baseline probability of membership in the engaged group (I), we assign Normal priors to the intercept γ_0 and the English language coefficient β_E:

$$\gamma_0 \sim N(\text{logit}(0.80), 0.25)$$

$$\beta_E \sim N(0, 0.3)$$

Model 2 Results

The model successfully converged, with all parameters showing $\hat{R} \approx 1.00$. A single divergent transition (0.025% of post-warmup draws) was observed, which is negligible. The fitted model results are summarized in the model output.

The second mixture model introduces demographic covariates (Academic Rank , Science Courses , and English) to predict the probability of membership in the engaged group (θ_2), while keeping the performance characteristics (μ, ϕ) of the two groups fixed across the population.

1. Shift in group definitions

A critical finding in model 2 is the substantial shift in the definition of the low engagement (**R**) group compared to model 1.

- Group **R**: The estimated mean μ_R increased to 0.57, corresponding to an average score of 14.28 out of 25 (57%).
- Group **I**: The estimated mean μ_I increased to 0.82, corresponding to an average score of 20.46 out of 25 (82%).

By allowing covariates to explain who belongs to which group, the model re-optimized the group definitions. The data strongly rejects the existence of a guessing cluster (SLCI Score Count ≈ 6). Instead, the model separates the population into average or inconsistent students (group **R**) and high performing students (group **I**). The low group is now scoring well above the midpoint of the exam, showing that pure random guessing is statistically negligible.

2. Determinants of group membership (θ_2)

The mixing model $\theta_{2,i} = \text{logit}^{-1}(\eta_i)$ identifies which factors predict belonging to the high performing (**I**) group.

A. English language effect: The fixed effect for English is strong and positive ($\beta_{\text{Yes}} = 1.18$).

- Non-native speakers: The baseline intercept (θ_2 intercept ≈ 0.12) implies a probability of $\text{logit}^{-1}(0.12) \approx 53\%$ of being in the high performing group.

- Native speakers: Adding the coefficient ($0.12 + 1.18 = 1.30$) implies a probability of $\text{logit}^{-1}(1.30) \approx 79\%$.
- Conclusion: Native English speakers are significantly more likely to belong to the high performing cluster than non-native speakers.

B. The Impact of science training: The random effect for Science Courses has a large standard deviation ($\sigma_{\text{sci}} = 1.18$).

- This variance is more than double the magnitude of the Academic Rank variance ($\sigma_{\text{year}} = 0.55$).
- Conclusion: The number of science courses a student has completed is the dominant predictor of their performance classification. Whether a student falls into group **I** depends much more on their specific science curriculum exposure than on their year.

3. Precision and consistency

- Group **I** ($\phi_I \approx 36$): The high performing group is very variable (overdispersed).
- Group **R** ($\phi_R \approx 7.8$): The lower performing group is much more consistent.
- This suggests that group **I** represents a relatively homogeneous population of high performing students, while group **R** reflects a more heterogeneous mix of partial knowledge and inconsistent engagement.

(6) Approach Three

While the previous approach made the reasonable assumption that the demographic covariates likely influence whether a student will be low engaged/guessing or not, our final approach expands the model further by considering how these demographic covariates effect the actual shape of the Beta distribution prior for p_i.

Model 3 Notation

Now, we allow per-student Beta shapes using $(\mu_{g,i}, \phi_{g,i})$:

$$p_i \mid (z_i = g) \sim \text{Beta}(\alpha_{g,i}, \beta_{g,i})$$
$$\alpha_{g,i} = \mu_{g,i}\phi_{g,i}, \quad \beta_{g,i} = (1 - \mu_{g,i})\phi_{g,i}$$

For low engagement (**R**), we keep it near guessing with a constant (and strongly regularized):

$$\text{logit}(\mu_{R,i}) = a_R, \quad \log(\phi_{R,i}) = c_R$$

For engaged (**I**), we then make covariates affect both mean and dispersion because within this group the actual experiences of the students matter:

$$\text{logit}(\mu_{I,i}) = a_I + u^{(I)}_{\text{year}[i]} + v^{(I)}_{\text{sci}[i]} + \beta^{(I)}_E \cdot 1\{\text{English}_i\} + b_{\text{inst}[i]}$$

$$\log(\phi_{I,i}) = c_I + u^{(\phi)}_{\text{year}[i]} + v^{(\phi)}_{\text{sci}[i]} + \beta^{(\phi)}_E \cdot 1\{\text{English}_i\}$$

The Institution effect is added to the engaged mean to capture performance differences across universities:

$$b_l \sim N(0, \sigma_{\text{inst}})$$

Additionally, the covariates are still able to affect mixing since the premise of the previous approach is still sound:

$$\theta_{2,i} = \text{logit}^{-1}(\gamma_0 + u_{\text{year}[i]} + v_{\text{sci}[i]} + \beta_E \cdot 1\{\text{English}_i = \text{No}\})$$

This approach reduces the possibility of low ability being mistaken for low engagement by having covariates push around engaged mean ($\mu_{I,i}$) and dispersion ($\phi_{I,i}$), so truly low ability but actually engaged students can more easily be included in group **I** if their demographic information supports their low ability, rather than forcing the **R** group, which is anchored near guessing, to inflate upward.

We assign the same weakly informative Exponential(1) priors to all group-level variance parameters (σ), and Normal priors to the fixed effects (a, c, β).

Model 3 Results

The model successfully converged, with all parameters showing $\hat{R} \approx 1.00$. Only 4 divergent transitions (0.1% of post-warmup draws) were observed, which is negligible. The fitted model results are summarized in the model output.

1. Resolution of the guessing group (**R**)

A crucial result of this final model is that it returns the low engagement group (**R**) to a definition consistent with random guessing or very low ability.

- Mean score: The estimated mean μ_R is 0.34, corresponding to an average score of 8.51 out of 25.
- Comparison: This is significantly lower than model 2 (which drifted up to ≈ 14) and much closer to the true random baseline of 6.25.
- Interpretation: By allowing the engaged group (**I**) to have variable performance (ie: allowing engaged students to be low scoring), the model no longer needs to force low performing students into the guessing bin. This isolates the true low engaged part of the population scoring barely above chance.

2. The structure of engagement (group **I**)

The engaged group now better represents the full spectrum of students who are trying, ranging from low ability to high ability.

- Baseline performance: The average engaged student (intercept) has a mean $\mu_I \approx 0.72$, or a score of 18.08 out of 25.
- Impact of English: Being a native English speaker increases the performance of engaged students significantly ($\beta_{\mu,\text{Yes}} = 0.20$), but has a negligible effect on their consistency ($\beta_{\phi,\text{Yes}} \approx -0.12$, with 95% CI including zero).

- Institutional variation: The random effect for Institution ($\sigma_{\text{inst}} = 0.38$) is more than double the effect of Academic Rank on performance ($\sigma_{\text{rank}} = 0.15$). This implies that which university a student attends has a larger impact on their engaged score than their year in school.

3. Drivers of group membership (θ_2)

Even after accounting for ability differences within the engaged group, demographic factors still strongly predict who falls into the low engagement group.

- Science Course ($\sigma_{\text{sci}} = 0.51$): This remains the strongest predictor of engagement. Students with more science background are consistently less likely to be classified in the low engagement group.
- English : Native speakers are more likely to be classified as engaged ($\beta_{\theta,\text{Yes}} = 0.32$). Although even for the baseline group, the intercept estimate of 1.34 shows that $\approx 79\%$ of students are classified as engaged ($\text{logit}^{-1}(1.34) \approx 0.79$). Adding the coefficient for English to the intercept ($1.34 + 0.32 = 1.66$) yields a probability of $\text{logit}^{-1}(1.66) \approx 84\%$.
- Academic Rank : School year plays a moderate role, but less so than specific coursework.

4. Benefits of model 3

Model 3 provides the most theoretically sound description of the data. It successfully:

- Reclaims the definition of group **R**: Identifying a small, distinct cluster of students scoring near chance (≈ 8.5).
- Validates the low ability student: Allowing engaged students to score lower (variable μ_I) prevents them from being misclassified as guessers.
- Quantifies institutional bias: Revealing significant performance differences across universities (or institutions), which was a sub-question for researchers.

(7) Model Comparisons

With our three models fitted, we can now compare using Leave-One-Out Cross-Validation and the Pearson chi-square statistic as measures of goodness of fit.

Cross-Validation

We first compare the predictive performance of the three fitted models using Leave-One-Out Cross-Validation (LOO-CV). The table below summarizes the Expected Log Predictive Density (ELPD) difference relative to the best performing model:

```
##        elpd_diff se_diff
## fit_m3    0.0       0.0
## fit_m2 -598.5      34.4
## fit_m1 -969.3      41.8
```

Model 3 is decisively the best model. The comparison shows a massive improvement in predictive power moving from model 2 to model 3 (elpd_diff $= -598.5$). Given the standard error of the difference (34.4), model 3 is ≈ 17 standard errors better than model 2 ($598.5/34.4 \approx 17.4$). In Bayesian model comparison, differences greater than 4 standard errors are generally considered robust evidence; a difference of 17 is very significant.

The superiority of model 3 over model 2 confirms that demographic factors (Academic Rank , Science Course , English) do not merely determine whether a student falls into the engaged group, they also significantly predict how well (μ_I) and how consistently (ϕ_I) that student performs once they are engaged.

Model 1 is by far the worst performing (elpd_diff $= -969.3$), confirming that a simple guesser and non-guesser mixture without demographic context is insufficient for describing student performance on the SLCI.

Posterior Predictive Checks Analysis

Figures 7, 8, and 9 show the posterior predictive checks for our three models. Posterior predictive checks assess whether a fitted model can plausibly generate data resembling the observed outcomes. In each figure, the dark line represents the observed distribution of scores, while each light line corresponds to one posterior predictive draw (with ndraws $= 50$).

For model 1, the observed distribution lies within the cloud of posterior predictive draws across most of the score range, indicating that the model captures the overall shape of the data reasonably well. Model 2 is by far the worst at matching the observed distribution's shape, specifically near the small guessing bump at SLCI Score Count 6.25. Finally, the posterior predictive draws for model 3 are similar to those in model 1, with both fitting the curve's shape well but slightly underestimating the high engagement peak near SLCI Score Count 20.

Pearson Chi-Square

The goodness of fit analysis using the Pearson chi-square statistic (χ^2) matches this result. While model 1 yields the lowest χ^2 value (50.5), this is likely because it fits the marginal distribution of scores without being constrained by individual student characteristics. It effectively curve fits the overall histogram shape perfectly but fails to understand which student belongs where.

In contrast, model 2 exhibits the poorest fit ($\chi^2 \approx 118.5$), suggesting that forcing group membership to depend on covariates while keeping the performance distributions fixed is problematic. The model predicts students should be engaged based on their demographics, but the rigid performance parameters fail to match their actual scores.

Model 3 resolves this problem, achieving a χ^2 (64.3) comparable to the unconstrained baseline. This shows that model 3 successfully balances reproducing the overall population shape with the massive improvement in individual level predictive accuracy shown by the LOO analysis. Note that caching was not working for this R chunk, so these chi-square statistics might differ from the given output.

(8) Conclusions

I have divided the conclusion into a recap of the results from the analysis and a second section that presents recommendations for any future iterations of this study.

Results

The analysis gives evidence of two distinct latent groups within the SLCI test-takers: a low engagement group (**R**) comprising $\approx 16\%$ of the population, and an engaged group (**I**) comprising the remaining 84%. While the theoretical random guessing score is 6.25 (25%), the low engagement group is best estimated to have a mean score of ≈ 8.5 (34%). Earlier iterations of the model (model 2) artificially inflated this group's mean to roughly 14.3 by forcing low ability students into the low engagement group. However, by allowing for variable performance among engaged students in the final model (model 3), we successfully isolated a distinct cluster of students scoring near chance, separating them from those who are genuinely attempting the test but lack specific content knowledge.

For the engaged majority (**I**), performance is highly heterogeneous and strongly predicted by demographic covariates rather than being just high achievers. The average engaged student scores ≈ 18.1 (72%), but this baseline shifts significantly based on background. The number of completed Science Courses emerged as the strongest predictor of engagement, while native English speakers demonstrated significantly higher performance scores than non-native speakers. Furthermore, the analysis revealed substantial institutional variation ($\sigma_{\text{inst}} = 0.38$), indicating that a student's university context exerts a stronger influence on their engaged performance score than their Academic Rank ($\sigma_{\text{rank}} = 0.15$).

Model comparisons decisively favor the fully heterogeneous approach (model 3) over simpler alternatives, with predictive accuracy improving by ≈ 17 standard errors over the previous mixture model. This improvement comes from the model's ability to distinguish between low engagement and low ability. While simpler models effectively curve-fit the aggregate population histogram, they failed to accurately predict individual student outcomes. The final analysis concludes that demographics do not merely determine whether a student is engaged, but shape the level and consistency of their performance, validating the low ability student as distinct from the random guesser.

A key limitation of this final model (and the other ones) is that it treats low engagement and random guessing as a single group, as the available data cannot reliably distinguish them without fitting to noise. This combined classification makes the subtraction plot that would be used to disprove the Dunning-Kruger effect largely invalid at the lower end of the score distribution. Since model 3 classifies nearly every student scoring below ≈ 8 as low engagement, removing this group creates instability that ruins any estimated effect for engaged students in this range. This apparent inseparability suggests that an important missing variable is needed to clarify the distinction between guessing and low ability, a possibility I explore in the next section.

Design Recommendations

One variable missing from the provided data is the time taken to complete the SLCI. This is likely due to Google Forms lacking a built-in feature to track time. However, this data is crucial, as it is probable that disengaged students or those guessing randomly take significantly less time to complete the form because they are not reading the questions.

If researchers wish to continue using Google Forms, the Quilgo add-on is a viable solution. It records start and finish times and tracks if students switch tabs or windows during the assessment. This last feature is particularly useful for identifying instances where students have a long test duration but low engagement. Without these metrics, there is a continued risk of over-analyzing the data while trying to differentiate between two latent groups that are currently difficult to distinguish.

Finally, I recommend adjusting the self-assessment checks to ask how many questions the student thinks they answered correctly out of 25, rather than asking for a percentage. It is easier for students to estimate a raw score than to perform the additional step of converting that number into a percentage.

Appendix

EDA Figures

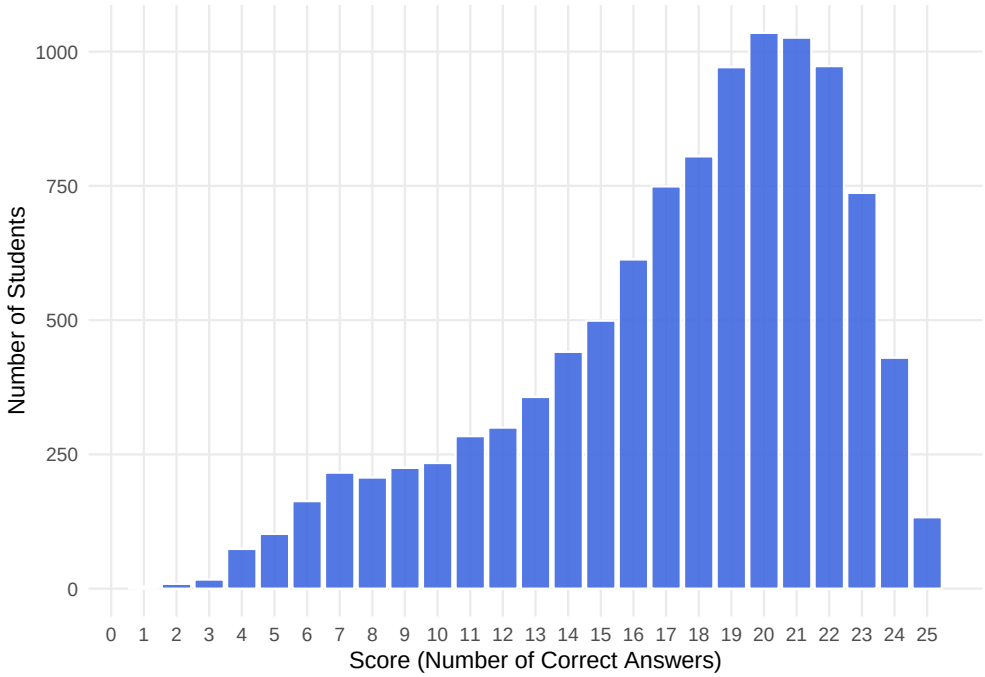

Figure 1: Distribution of SLCI Scores

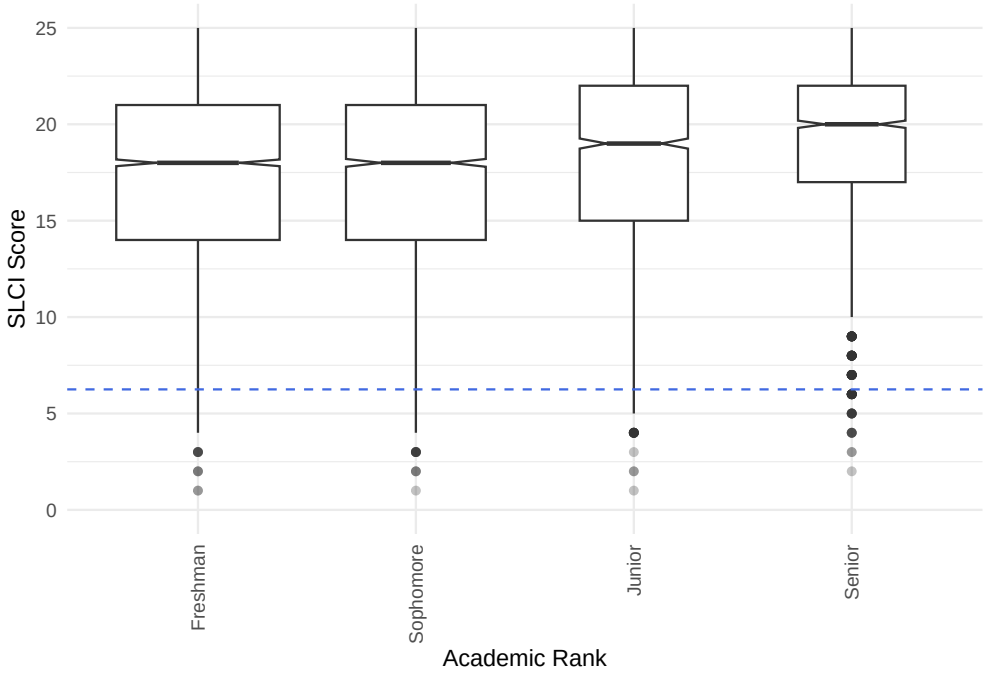

Figure 2: SLCI Scores by Academic Rank

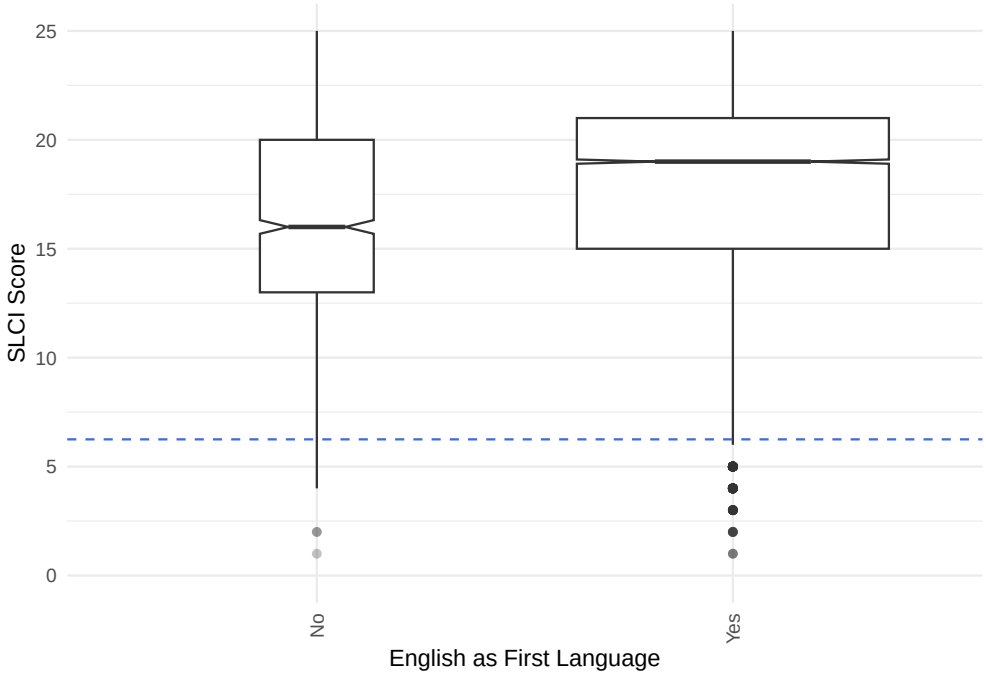

Figure 3: SLCI Scores by Language

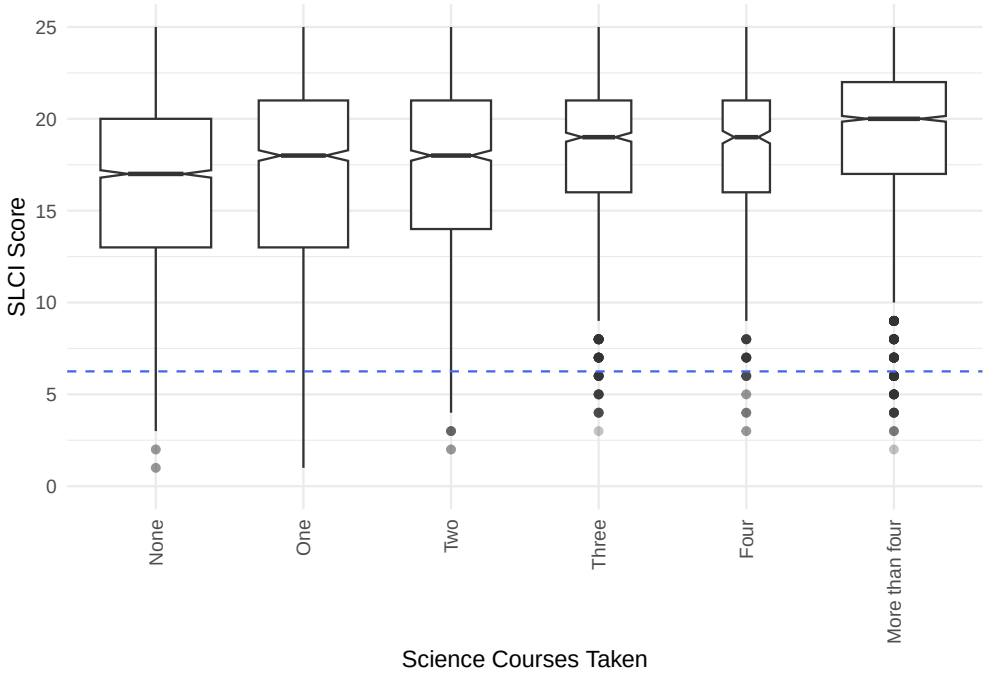

Figure 4: SLCI Scores by Science Courses

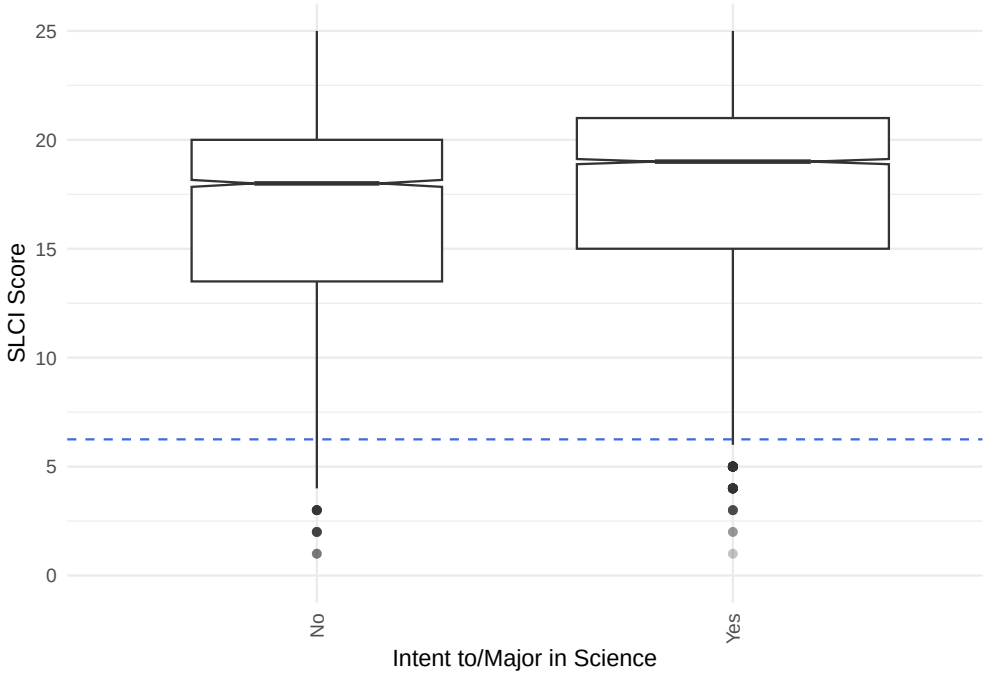

Figure 5: SLCI Scores by Science Intent

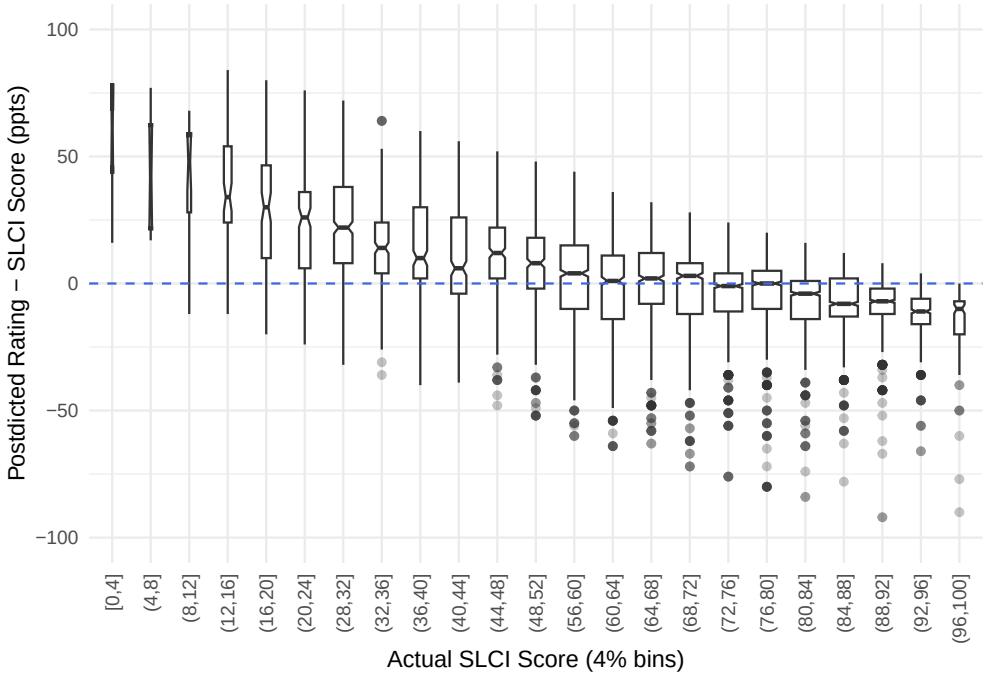

Figure 6: Self-Assessment Inaccuracy by SLCI Score

Posterior Predictive Checks

Model 1

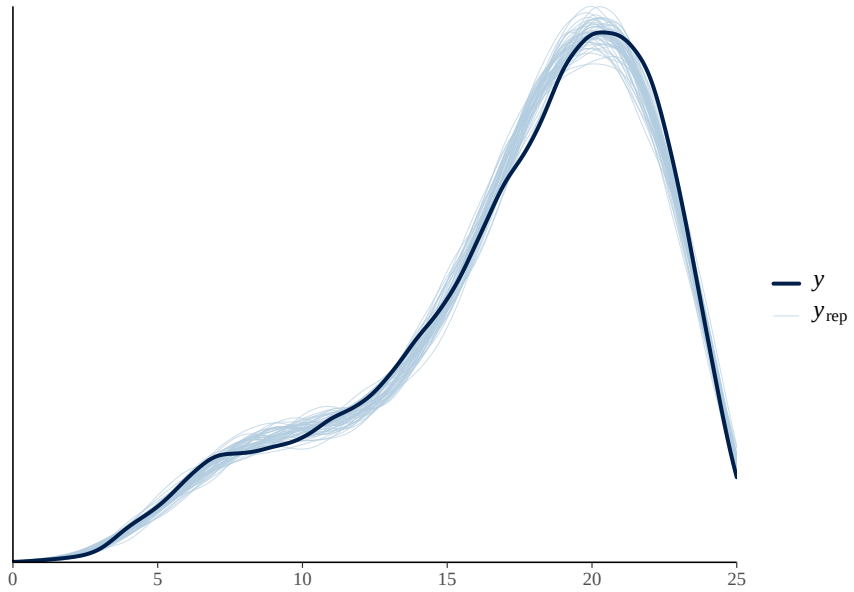

Figure 7: Model 1 Posterior Predictive Check

Model 2

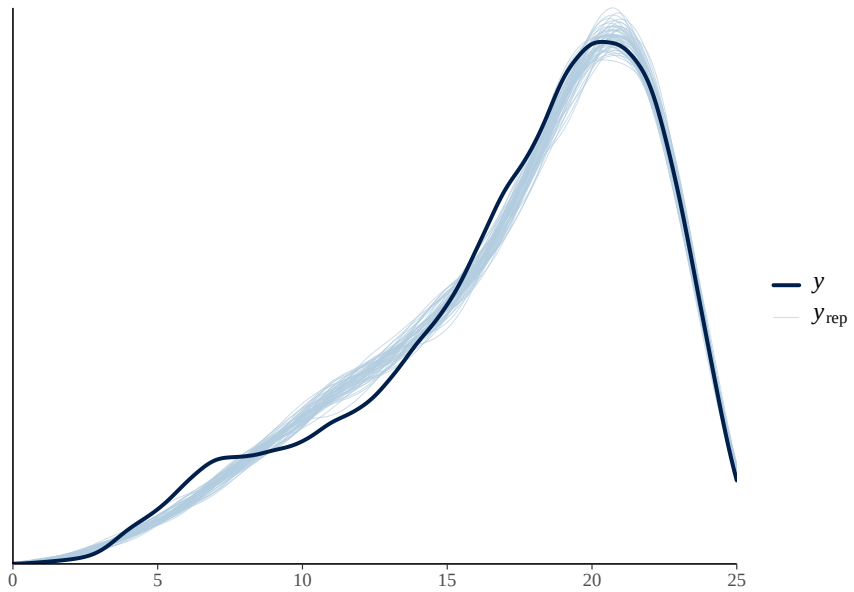

Figure 8: Model 2 Posterior Predictive Check

Model 3

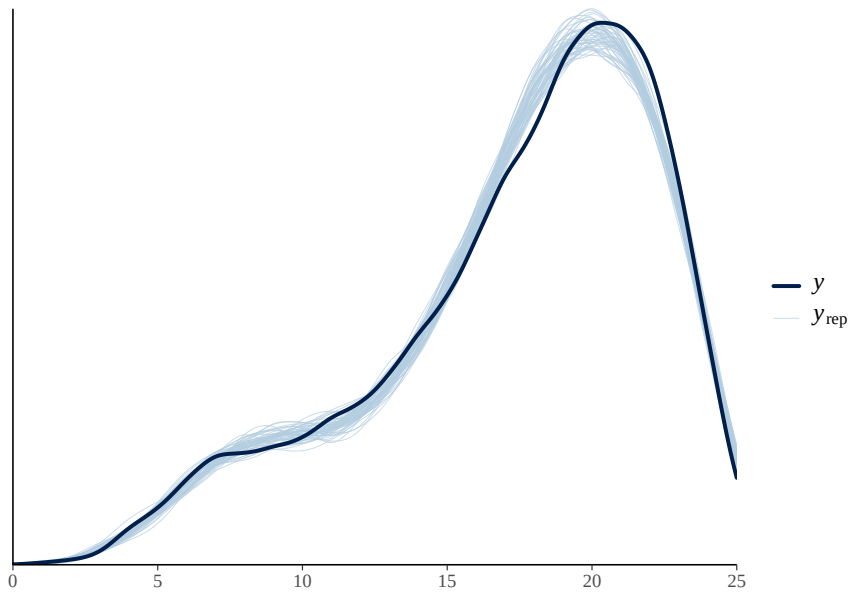

Figure 9: Model 3 Posterior Predictive Check

Model Summaries

Model 1 Summary

```
##  Family: mixture(beta_binomial, beta_binomial)
##   Links: mu1 = logit; mu2 = logit; phi1 = log; phi2 = log; theta2 = identit
## Formula: Score_Count | trials(25) ~ 1
##          mu1 ~ 1
##          phi1 ~ 1
##          mu2 ~ 1
##          phi2 ~ 1
##          theta2 ~ 1
##    Data: model_data (Number of observations: 10604)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Regression Coefficients:
##                  Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ES
## mu1_Intercept       -0.60      0.06    -0.71    -0.48 1.00      609      85
## phi1_Intercept       3.84      0.33     3.29     4.63 1.00      684      93
## mu2_Intercept        1.12      0.02     1.08     1.17 1.00      744     103
## phi2_Intercept       2.66      0.06     2.54     2.78 1.00      699     104
## theta2_Intercept     1.74      0.11     1.52     1.95 1.00      651      88
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

Model 2 Summary

```
##  Family: mixture(beta_binomial, beta_binomial)
##   Links: mu1 = logit; mu2 = logit; phi1 = log; phi2 = log; theta2 = identit
## Formula: Score_Count | trials(25) ~ 1
##          mu1 ~ 1
##          phi1 ~ 1
##          mu2 ~ 1
##          phi2 ~ 1
##          theta2 ~ 1 + (1 | Rank) + (1 | Science_Courses) + English
##    Data: model_data (Number of observations: 10604)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~Rank (Number of levels: 4)
##                      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## sd(theta2_Intercept)     0.55      0.42     0.15     1.71 1.01      897
##                      Tail_ESS
## sd(theta2_Intercept)     1600
##
## ~Science_Courses (Number of levels: 6)
##                      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## sd(theta2_Intercept)     1.18      0.45     0.56     2.33 1.01     1001
##                      Tail_ESS
## sd(theta2_Intercept)     1375
##
## Regression Coefficients:
##                   Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_E
## mu1_Intercept         0.29      0.03     0.22     0.35 1.00     2314     21
## phi1_Intercept        2.05      0.04     1.97     2.14 1.00     2696     27
## mu2_Intercept         1.51      0.02     1.46     1.55 1.00     3796     33
## phi2_Intercept        3.58      0.09     3.41     3.76 1.00     3961     23
## theta2_Intercept      0.12      0.27    -0.40     0.65 1.00     2873     25
## theta2_EnglishYes     1.18      0.12     0.95     1.42 1.00     3918     31
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

Model 3 Summary

```
##  Family: mixture(beta_binomial, beta_binomial)
##   Links: mu1 = logit; mu2 = logit; phi1 = log; phi2 = log; theta2 = identit
## Formula: Score_Count | trials(25) ~ 1
##          mu1 ~ 1
##          phi1 ~ 1
##          mu2 ~ 1 + (1 | Rank) + (1 | Science_Courses) + English + (1 | Inst
```

```
##          phi2 ~ 1 + (1 | Rank) + (1 | Science_Courses) + English
##          theta2 ~ 1 + (1 | Rank) + (1 | Science_Courses) + English
##    Data: model_data (Number of observations: 10604)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~Institution (Number of levels: 26)
##                   Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_E
## sd(mu2_Intercept)     0.38      0.06     0.28     0.53 1.00      791     16
##
## ~Rank (Number of levels: 4)
##                      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## sd(mu2_Intercept)        0.15      0.12     0.05     0.49 1.00     1311
## sd(phi2_Intercept)       0.19      0.16     0.01     0.63 1.00     1213
## sd(theta2_Intercept)     0.40      0.24     0.14     1.06 1.00     1529
##                      Tail_ESS
## sd(mu2_Intercept)        1428
## sd(phi2_Intercept)       1539
## sd(theta2_Intercept)     2446
##
## ~Science_Courses (Number of levels: 6)
##                      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## sd(mu2_Intercept)        0.14      0.06     0.06     0.30 1.00     1413
## sd(phi2_Intercept)       0.13      0.09     0.01     0.33 1.00     1169
## sd(theta2_Intercept)     0.51      0.22     0.25     1.05 1.00     1247
##                      Tail_ESS
## sd(mu2_Intercept)        2169
## sd(phi2_Intercept)       1027
## sd(theta2_Intercept)     2041
##
## Regression Coefficients:
##                   Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_E
## mu1_Intercept        -0.66      0.04    -0.73    -0.59 1.00     2652     26
## phi1_Intercept        4.18      0.33     3.64     4.94 1.00     2641     23
## mu2_Intercept         0.96      0.13     0.70     1.23 1.00     1008     15
## phi2_Intercept        3.09      0.18     2.75     3.46 1.00     2285     21
## theta2_Intercept      1.34      0.23     0.85     1.77 1.00     2298     26
## mu2_EnglishYes        0.20      0.03     0.15     0.26 1.00     4078     33
## phi2_EnglishYes      -0.12      0.12    -0.36     0.11 1.00     3446     27
## theta2_EnglishYes     0.32      0.11     0.10     0.54 1.00     4195     27
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

Alternative Goodness of Fit

```
##                                Model Chi_Square
## 1                Model 1 (Baseline)   53.56915
## 2 Model 2 (Covariates on Mixing)  116.65954
## 3       Model 3 (Full Covariates)   64.53641
```

Subtraction Plot

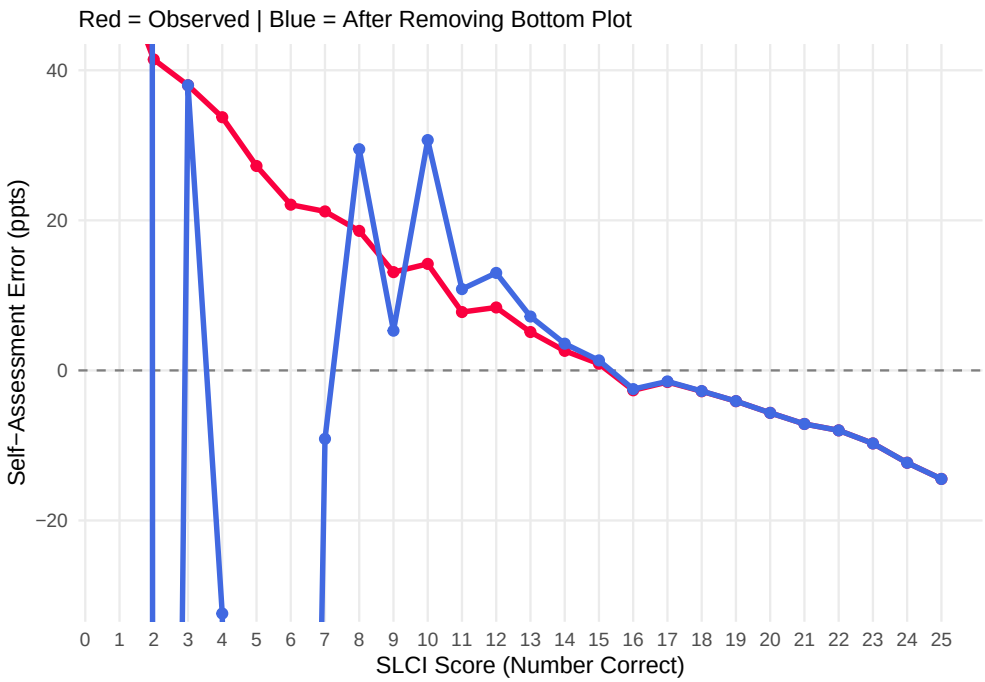

Figure 10: Removing Group R